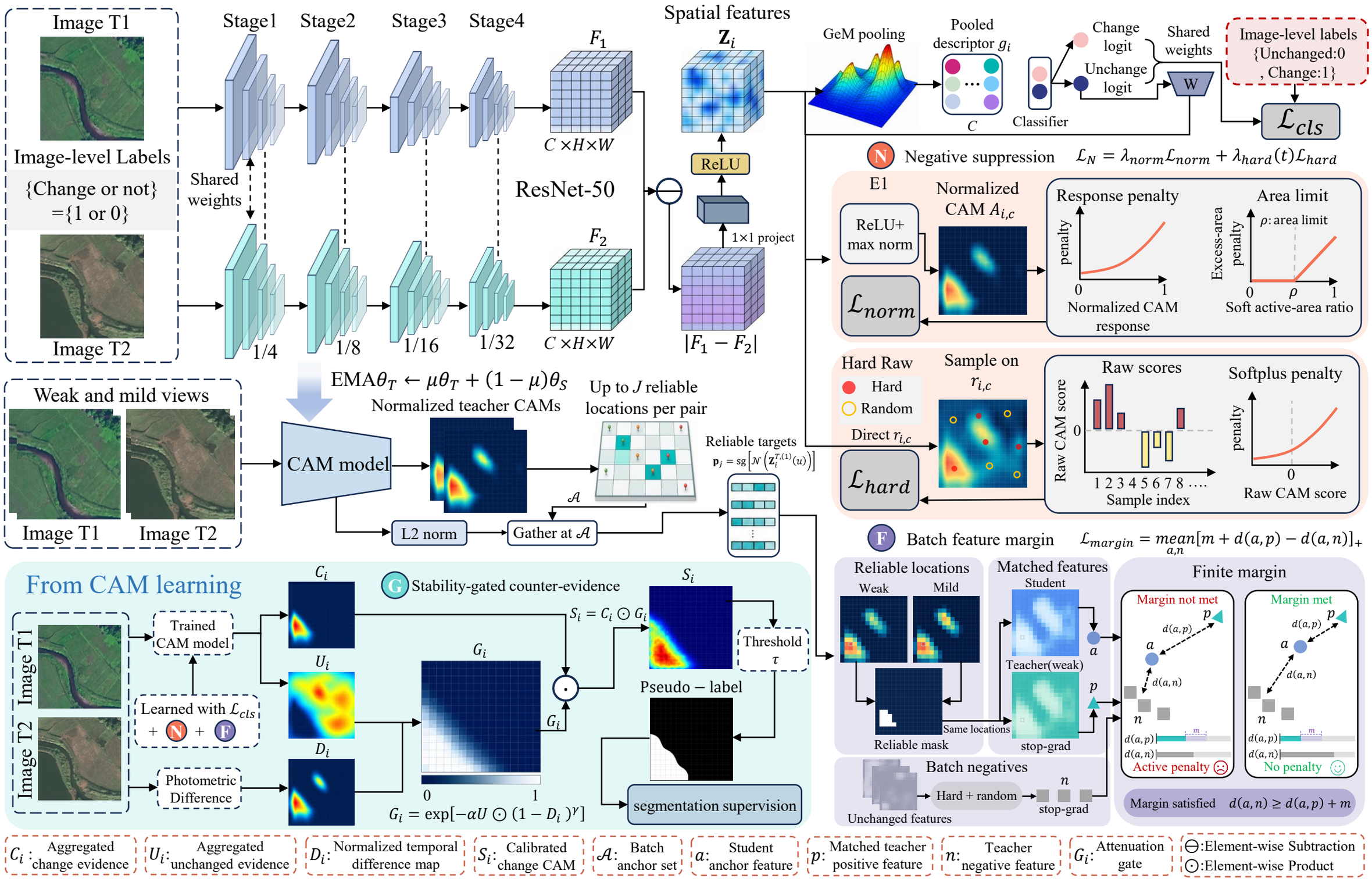

Image T1
Image-level Labels
{Change or not}
={1 or 0}
Image T2
Stage1
Stage2
Stage3
Stage4
Shared weights
1/4
1/8
1/16
1/32
ResNet-50
F_1
F_2
$C \times H \times W$
Spatial features
$\mathbf{Z}_i$
ReLU
1×1 project
$|F_1 - F_2|$
GeM pooling
Pooled descriptor g_i
C
Classifier
Change logit
Unchange logit
Shared weights
W
Image-level labels {Unchanged:0 , Change:1}
$\mathcal{L}_{cls}$
N Negative suppression
$\mathcal{L}_N = \lambda_{norm}\mathcal{L}_{norm} + \lambda_{hard}(t)\mathcal{L}_{hard}$
E1
ReLU+ max norm
$\mathcal{L}_{norm}$
Normalized CAM $A_{i,c}$
Response penalty
penalty
Normalized CAM response
Area limit
Excess-area penalty
ρ: area limit
Soft active-area ratio
Hard Raw
Hard
Random
Direct $r_{i,c}$
$\mathcal{L}_{hard}$
Sample on $r_{i,c}$
Raw scores
Raw CAM score
Sample index
Softplus penalty
Raw CAM score
F Batch feature margin
$\mathcal{L}_{margin} = \underset{a,n}{mean}[m + d(a,p) - d(a,n)]_+$
Weak and mild views
Image T1
Image T2
CAM model
$\text{EMA}\theta_T \leftarrow \mu\theta_T + (1-\mu)\theta_S$
Normalized teacher CAMs
L2 norm
Up to J reliable locations per pair
Gather at $\mathcal{A}$
Reliable targets
$\mathbf{p}_j = \text{sg}\left[\mathcal{N}\left(\mathbf{Z}_i^{T,(1)}(u)\right)\right]$
From CAM learning
Image T1
Image T2
Trained CAM model
Learned with $\mathcal{L}_{cls}$ + N + F
Photometric Difference
C_i
U_i
D_i
G Stability-gated counter-evidence
$S_i = C_i \odot G_i$
G_i
Threshold τ
Pseudo − label
segmentation supervision
$G_i = \exp[-\alpha U \odot (1 - D_i)^\gamma]$
Reliable locations
Weak
Mild
Reliable mask
Same locations
Matched features
Student
Teacher(weak)
stop-grad
Batch negatives
Hard + random
stop-grad
Unchanged features
Finite margin
Margin not met
Active penalty
Margin met
No penalty
Margin satisfied $d(a,n) \geq d(a,p) + m$
C_i: Aggregated change evidence
U_i: Aggregated unchanged evidence
D_i: Normalized temporal difference map
S_i: Calibrated change CAM
$\mathcal{A}$: Batch anchor set
a: Student anchor feature
p: Matched teacher positive feature
n: Teacher negative feature
G_i: Attenuation gate
⊖:Element-wise Subtraction
⊙:Element-wise Product